Parallel Expectation-Maximization for Clustering

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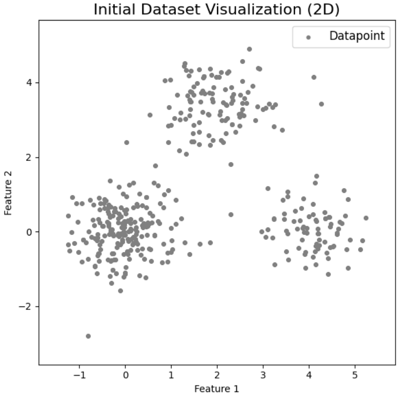
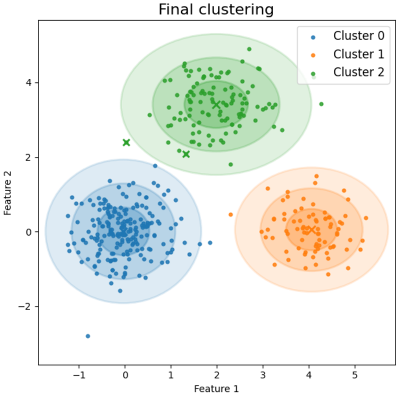
Contents

What is clustering - A Visual Example

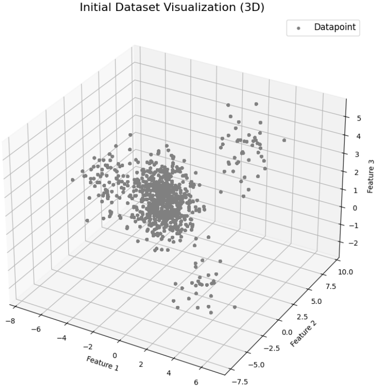


First iteration of the clustering process

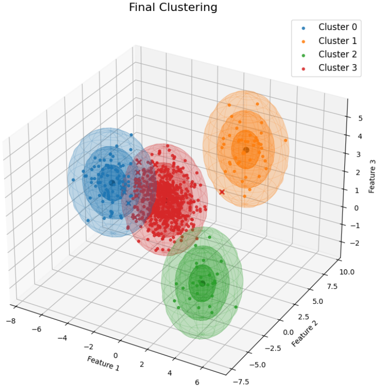


Final clustering result

What is clustering - A Visual Example (3D)



First iteration (3D view)



Final clustering (3D view)

What is clustering - More Formally

- Partition a dataset $\mathcal{X} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N\}$ where $\boldsymbol{x}_i \in \mathbb{R}^D$ into K groups (clusters).
- Each cluster is represented by a probability distribution with parameters θ_k.
- Find the cluster parameters that maximize the log-likelihood of the data:

$$\max_{\theta}(\mathcal{L}(\theta|\mathcal{X}))$$

$$\mathcal{L}(\theta|\mathcal{X}) = \sum_{i=1}^{N} \log\left(\sum_{k=1}^{K} \pi_k \, p(\boldsymbol{x}_i \mid \theta_k)\right)$$

Gaussian Mixture Models

- Each cluster is represented by a Gaussian distribution with its own mean and covariance:

$$p(\boldsymbol{x} \mid \theta) = \sum_{k=1}^{K} \pi_k \, \mathcal{N}(\boldsymbol{x} \mid \boldsymbol{\mu}_k, \Sigma_k)$$

$$\sum_{k=1}^{K} \pi_k = 1, \quad \pi_k \geq 0$$

- Thelikelihood of a data point $\boldsymbol{x}$ to belong to cluster k is given by:

$$\mathcal{N}(\boldsymbol{x} \mid \boldsymbol{\mu}_k, \Sigma_k) = \frac{1}{(2\pi)^{D/2} |\Sigma_k|^{1/2}} \exp\left(-\frac{1}{2} (\boldsymbol{x} - \boldsymbol{\mu}_k)^T \Sigma_k^{-1} (\boldsymbol{x} - \boldsymbol{\mu}_k) \right)$$

Gaussian Mixture Models – Approximation

- We assumed the features are uncorrelated within each component, simplifying the covariance matrix to a diagonal form:

$$\Sigma_k = \begin{pmatrix} \sigma_{k,1} & 0 & \cdots & 0 \\ 0 & \sigma_{k,2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_{k,D} \end{pmatrix}$$

- This implies that the likelihood can be expressed as:

$$\mathcal{N}(\boldsymbol{x} \mid \boldsymbol{\mu}_k, \Sigma_k) = \frac{1}{(2\pi)^{D/2} \prod_{d=1}^{D} \sigma_{k,d}^{1/2}} \exp\left(-\frac{1}{2} \sum_{d=1}^{D} \frac{(x_d - \mu_{k,d})^2}{\sigma_{k,d}} \right)$$

Chicken and Egg Problem

- To cluster the data, we need to know the parameters of the Gaussian components.
- To estimate the parameters, we need to know the cluster assignments of the data points.
- This circular dependency is known as the "chicken and egg" problem in clustering.

The solution - EM-Algorithm

The Expectation-Maximization (EM) algorithm is an iterative method to find maximum likelihood estimates of parameters in models with latent variables.

EM-Algorithm - Steps

The EM algorithm consists of two main steps:

- **E-step (Expectation step):** Compute the expected value of the latent variables given the current parameter estimates (soft assignments).
- **M-step (Maximization step):** Update the parameters to maximize the expected log-likelihood found in the E-step.

EM-Algorithm - Pseudocode

Algorithm 1 Expectation-Maximization for GMM clustering

```
Randomly initialize parameters {π_k, μ_k, Σ_k}_{k=1}^K
for i = 1 ... MAX_ITER do
    E-step: compute responsibilities
    M-step: update parameters
    Compute log-likelihood 𝓛
    if Converged then
        break
    end if
end for
Clustering: assign labels
```

EM-Algorithm - E-Step

Using the current parameter estimates θ, we compute the responsibilities, i.e. the posterior probability that data point i belongs to Gaussian component k.

$$\gamma_{ik} = \frac{\pi_k \, \mathcal{N}(\boldsymbol{x}_i \mid \boldsymbol{\mu}_k, \Sigma_k)}{\sum_{j=1}^{K} \pi_j \, \mathcal{N}(\boldsymbol{x}_i \mid \boldsymbol{\mu}_j, \Sigma_j)}$$

EM-Algorithm - M-Step

Using the responsibilities γ_{ik} computed in the E-step, we update the parameters:

- **Sum of responsibilities for each component:** $N_k = \sum_{i=1}^{N} \gamma_{ik}$
- **Update of means**: $\boldsymbol{\mu}_k = \frac{1}{N_k} \sum_{i=1}^{N} \gamma_{ik} \boldsymbol{x}_i$
- **Update of covariances**: $\Sigma_k = \frac{1}{N_k} \sum_{i=1}^{N} \gamma_{ik} (\boldsymbol{x}_i - \boldsymbol{\mu}_k)(\boldsymbol{x}_i - \boldsymbol{\mu}_k)^T$
- **Update of mixture weights**: $\pi_k = \frac{N_k}{N}$

Convergence

The log-likelihood of the data given the model parameters is computed as:

$$\mathcal{L}(\theta) = \sum_{i=1}^{N} \log \left(\sum_{k=1}^{K} \pi_k \, \mathcal{N}(\boldsymbol{x}_i \mid \boldsymbol{\mu}_k, \Sigma_k) \right)$$

Convergence it's reached if the change is below a predefined threshold ϵ.

Clustering Assignment

After convergence, each data point is assigned to the cluster with the highest responsibility:

$$\hat{z}_i = \arg\max_k p(z_i = k \mid \boldsymbol{x}_i, \theta) = \arg\max_k \gamma_{ik}$$

Computational Complexity Analysis

- The computational cost is expressed in terms of number of data points N, clusters K, and feature dimensions D.
- Both E-step and M-step involve operations over all data points and clusters:

$$O(NKD)$$

- Overall iteration cost:

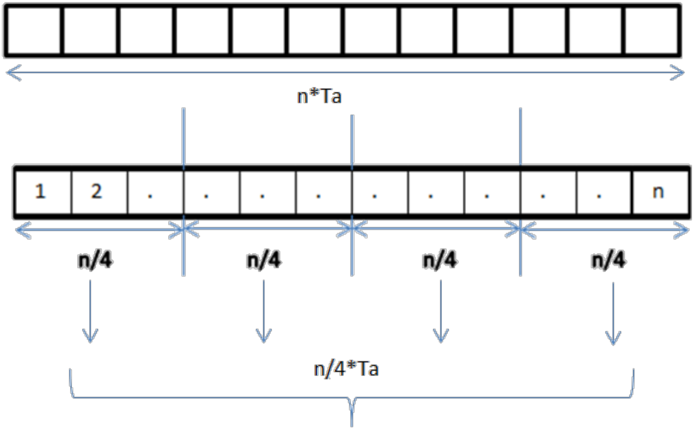
$$O(NKD) + O(NKD) = O(NKD)$$

Parallel Design - Literature Review

State-of-the-Art Approaches

- *Massively parallel expectation maximization using graphics processing units*, M. C. Altinigneli, C. Plant, and C. Bohm
- *A simple parallel em algorithm for statistical learning via mixture models*, S. X. Lee, K. L. Leemaqz, and G. J. McLachlan

Our Approach - Data Parallelization Strategy



Our Approach - Data Distribution

Our Approach - E-Step

- Each worker computes the responsibilities for its local subset of data points;
- Given the parameters θ, the responsibilities are computed independently for each data point
- No inter-worker communication is required during this phase.

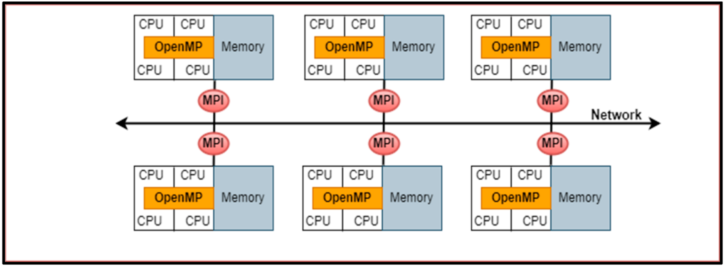
Our Approach - M-Step

- Each worker computes local sufficient statistics based on the soft assignments calculated in the E-step;
- These local statistics are then aggregated to update the global model parameters;
- Each worker will receive the updated parameters for the next E-step.

Performance Considerations

- Theoretically, the data parallelization strategy should decrease the complexity to;
$$O\left(\frac{N \cdot K \cdot D}{P}\right)$$
where P is the number of workers;
- Usually, real-world datasets have $N \gg K$ and $N \gg D$, making data parallelization more effective;
- The communication overhead during the M-step aggregation scales with K and D. This may become a bottleneck with large datasets.

Hybrid Parallelization



Hybrid Parallelization - Theoretical Speedup Analysis

- Reduces the number of MPI processes, lowering the number of communication events and overhead;
- Each MPI Process works on a larger subset of data, allowing computation to occur within a shared-memory environment;

Other Parallelization Strategies

- **Task Parallelization**
- **Pipeline Parallelization**
- **Model Parallelization**

Implementation

Scalability

Conclusion

Thank you!

[1] A. P. Dempster, N. M. Laird, and D. B. Rubin, "Maximum likelihood from incomplete data via the em algorithm," *Journal of the Royal Statistical Society: Series B*, vol. 39, no. 1, pp. 1–38, 1977.

[2] D. Reynolds, *Gaussian Mixture Models*. Boston, MA: Springer US, 2009.

[3] C. M. Bishop, *Pattern Recognition and Machine Learning*. Springer, 2006.

[4] A. Kak, "Expectation–maximization algorithm for clustering multidimensional numerical data: An rvl tutorial," Tech. Rep. —, Purdue University, March 3 2024.

[5] M. C. Altinigneli, C. Plant, and C. Böhm, "Massively parallel expectation maximization using graphics processing units," in *Proceedings of the 19th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, KDD '13, (New York, NY, USA), p. 838–846, Association for Computing Machinery, 2013.

[6] S. X. Lee, K. L. Leemaqz, and G. J. McLachlan, "A simple parallel em algorithm for statistical learning via mixture models," in *2016 International Conference on Digital Image Computing: Techniques and Applications (DICTA)*, pp. 1–8, 2016.

[7] I. Azizi, "Parallelization in python - an expectation-maximization application," tech. rep., Université de Lausanne, Département des Opérations (DO), June 2023. Inspired by CUSO Informatique 2023 Winter School.